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Mid-Term Examination March - 2025

Subject Name/ Code: CSE/IT -222

MM:20

Time Allotted: 90 min

NOTE: 1. All questions are compulsory.
2. Mark answer number clearly. Answers without precision receive NO credit.

1. A	Compare different cache mapping techniques (Direct-Mapped, Fully Associative, and Set-Associative) in terms of conflict misses when mapping data blocks from main memory to cache. Analyze and Compute Best and Worst-case probability of conflict misses for each technique. Based on your analysis, suggest the most efficient technique to minimize conflict misses while considering practical implementation constraints.	3
B.	Explain the statement that pipelining is used to reduce the average instruction processing time rather than the execution time of a single instruction.	2
2. A	Consider a cache memory with 80% Hit for Read operation and 90% HIT for write operation. The cache access time is 5 times faster than main memory access time, which is 20ms. If Write-back updation is used then if there is a Miss for read or write, a 2-word block (1 block = 1 word) must be moved from main memory to cache memory. Assume that cache is initially full and at any point of time 20% of its block have modified. Compute Avg. Read access time and Avg. write access time.	3
B.	Discuss the consequences of Cache Coherence problem and provide a comparative analysis of the techniques used to address the issue.	2
3. A	Consider a hypothetical system that supports one-address, two-address, and three-address instructions. The system uses fixed-length 24-bit instructions stored in a 256-word memory. If the system is using 2 two-address instructions and 3 three-address instructions, how many one-address instructions can be supported by the system?	3
B.	Consider a 4-block cache with 2-way Set-Associate mapping using the LRU as block replacement policy. What will be the minimum no. of Miss references if the following MM blocks are used by the processor: 8, 12, 0, 12, 8	2

4. A.	Discuss how operand forwarding technique is useful in reducing Stall Penalty in Pipeline processing. Then consider a five-stage instruction Pipeline with instruction fetch, instruction decode, operand fetch, perform operation and write operand stages. All the stages except perform operation (PO) consume 1-clock/instruction/stage. The perform operation requires 6 clocks for Division, 3 Clocks for Multiplication and 1 clock each for Addition and Subtraction. How many minimum clocks are required to implement the following four instructions with operand forwarding. I0: MUL R2 R0 R1 I1: DIV R5 R3 R4 I2: ADD R2 R5 R2 I3: SUB R5 R2 R6	3
B.	If it is required to implement a 4-Stage Pipeline such that Stage-2 delay is twice the Stage-1 delay and half of Stage-3 delay. Stage-4 and Stage-2 delay are same. The Stage-3 delay is 40ns, then how much time is required to complete 100 instructions and what efficiency can be achieved with this Pipeline?	2